

Problem 1: Applying Scrum in a Telemedicine App Project

1. Explanation of Scrum Roles and Responsibilities

- **Product Owner (PO):** The single voice of the customer and stakeholders. The PO is responsible for maximizing product value, managing and prioritizing the Product Backlog, and clearly articulating requirements through user stories and acceptance criteria.
- **Scrum Master (SM):** A servant-leader dedicated to the Scrum team and organization. The SM facilitates Scrum ceremonies, coaches the team in self-organization and cross-functionality, and removes external impediments blocking progress.
- **Development Team:** A self-organizing, cross-functional group of professionals (developers, QA engineers, UX/UI designers) who perform the work to deliver a potentially releasable increment of "Done" software at the end of each Sprint.

2. Root Cause Analysis

- **Lack of Upfront Backlog Grooming:** Requirements are being introduced dynamically during Sprint Planning because the backlog has not been analyzed, sized, and refined prior to the meeting.
- **Stakeholder Bypass:** Stakeholders and executives are directly approaching developers or the Product Owner mid-sprint to inject urgent changes, bypassing formal change-management channels.
- **Ambiguous Product Vision:** Without a clearly defined Minimum Viable Product (MVP) scope, the PO lacks firm boundaries to reject low-priority or out-of-scope requests.
- **Absence of a Sprint Goal Freeze:** The team is treating the sprint backlog as mutable, violating the core Agile principle of Sprint Stability.

3. Improved Backlog Prioritization Plan

- **MoSCoW Method:** Categorize backlog items into *Must have*s, *Should have*s, *Could have*s, and *Won't have this time*. Only *Must have*s enter the upcoming Sprint Backlog.
- **Value vs. Effort Matrix:** Evaluate user stories based on business value (revenue, user retention, compliance) versus technical complexity to prioritize high-impact, low-effort items first.
- **Refinement Cadence:** Dedicate 10% of every sprint to a formal **Backlog Refinement (Grooming) session** where stories for the next two sprints are estimated, clarified, and decomposed.
- **Sprint Scope Lock:** Enforce a strict policy that once a Sprint Planning meeting concludes and the Sprint Goal is set, no new features can be added. Incoming urgent requests must go into the product backlog for the subsequent sprint.

4. Retrospective Improvement Suggestions

- **Establish a Definition of Ready (DoR):** Ensure no user story enters a sprint unless it has clear acceptance criteria, UI/UX wireframes, and team estimation.
- **Stakeholder Communication Protocol:** Direct all stakeholder feedback through structured review sessions (Sprint Reviews) rather than ad-hoc interruptions..

